Venkatesh Manipati

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About

Full Stack Developer and AI/ML enthusiast with strong expertise in Python, React, OpenCV, and deep learning. Experienced in building production-ready web applications, AI pipelines, and research-driven solutions. Skilled in data preprocessing, model training, automation, and cloud-based deployment. Passionate about leveraging technology to solve real-world problems efficiently.

Work Experience

Smart-India-Hackathon *Sep 2025 – Oct 2025*

I worked on building an AI-driven train traffic control system that helps manage schedules more efficiently and reduces clashes between trains sharing the same tracks. By using real-time data such as train location, speed, and delays, I created a decision model that improves routing and cuts down unnecessary waiting time at stations. Overall, the project focused on making railway operations smoother and more efficient through smart use of machine learning and traffic management logic.

Salesforce Developer Intern *May 2025 – Jul 2025*

During my internship, I gained hands-on experience working with Salesforce CRM, where I explored how real-world business processes are managed on the platform. I learned and practiced Apex programming and built user interfaces using Lightning Web Components (LWC). I also worked on implementing basic automation by writing Apex triggers and using SOQL queries to handle business logic and data operations efficiently.

Web Development Intern – SRM AP *Jul 2024 – Apr 2025*

I developed a full-stack stationery platform that digitized the campus printing process, helping students place orders online and significantly reducing waiting time. To make pricing accurate and efficient, I built a Python-based automated page counter that instantly calculated document pages and costs, minimizing manual errors. I also integrated the Razorpay payment gateway and designed a real-time admin dashboard, enabling smooth payment processing, order tracking, and status updates.

Projects

Rail-AI-Control Git Repository

Python, FastAPI, Docker, Next.js, Tailwind CSS, Bootstrap

Developed a predictive analytics module to detect potential train congestion in advance. The system analyzed traffic patterns and operational data to anticipate bottlenecks before they occurred. Based on these insights, it suggested smarter rerouting strategies to ensure smoother and more efficient railway operations.

SRMAP Stationery Hub */ Node.js, Thress.js, MongoDB, Tailwind CSS* Live Demo | Git Repository

Designed and developed a comprehensive web-based platform to digitize and streamline the university stationery shop's printing workflow, reducing student wait times and optimizing order management.

LocalLens - Smart Business Discovery Git Repository

Python, Flask, Google Places API, Google Sheets API, HTML, Jinja2, REST APIs, CSV Handling

Developed a Flask-based business search web app that integrates Google Places API, previews results in-browser, and supports CSV export and Google Sheets automation.

Nexi - SmartVoice Assistant Git Repository

Python, SpeechRecognition, PyAudio, pyttsx3, OpenAI API, NLTK, Flask, OS, Web Automation Libraries

Built a smart voice assistant in Python that listens to spoken commands, understands them, and responds using text-to-speech while performing tasks like opening apps and searching the web. This project gave me hands-on experience in speech recognition, automation, and building interactive AI-driven systems.

Online Food Ordering System Git Repository

C++, Object Oriented Programming

Created two types of users: Manager and customer. Manager can choose to do any of the following: 1.View Menu 2.Update Menu Item 3.Update Price 4.View Total Sales 5.View Popular Dish. Customer can do any of the following: 1.View Menu 2.Place Order 3.Display Ordered Items

Education

2023 - 2027 Bachelor's Degree, SRM-AP University (CGPA: 7.97/10)
2022 Class 12, SSC Board (GPA: 8.5/10)
2021 Class 10, SSC Board (GPA: 10/10)

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript, Java, HTML/CSS, SQL, PHP
Frameworks: React.js, Flask, Bootstrap, MongoDB
Developer Tools: Git, VS Code, Google Colab, Jupyter Notebook, Docker, Google Antigravity
Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, OpenCV, MediaPipe, cvzone, Requests, BeautifulSoup
Currently Learning: Deep Learning, Express.js, Computer Networks (CN), Automata and Compiler Design, NLP

Certifications

Coursera: Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders, Reinforcement Learning.
Oracle: Java SE 17 fundamentals, OOP, Java Collections, Java Stream API, Java Concurrency.
MongoDB: 2026 MongoDB Associate Developer